

Attack–Fault Identifiability and the Cost of Ambiguity at the Electronic Trading Gateway

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Abstract

Operators of electronic trading gateways must respond to suspicious order flow before knowing whether a cyber compromise or a software fault caused it. For a linear-Gaussian model of gateway-observable order flow, we prove that a compromise and a fault are observationally indistinguishable if and only if their projected injections coincide, and that under this confounding no policy using the observation beats the best constant action, so ambiguity has a strictly positive cost, bounded below also under approximate confounding. If the incident propagates through the dynamics instead of shifting the observation only, confounding survives only when the injection difference lies in the Kalman unobservable subspace, which fails for generic dynamics. We then build, in the ABIDES agent-based market simulator, injection designs that match compromise and fault on chosen summary statistics; this is not a test of the theorem, whose premise the constructions do not meet, and diagnosis uses complete episodes. Calibration by search did not bring the summary-matched regime to chance: every summary-matched accuracy interval lies above chance (lowest bound 0.525, in Study 5), and in some designs it is more separable than the contrast regime. With a positive fault loss, a classifier-based policy has a lower point-estimate cost than always-contain in all 50 cells (a supported improvement in 49) and is never worse than the best constant policy with a supported interval; cells without a supported improvement lie at low priors or in one profile. Costs are an illustrative market-impact reconstruction, not measured losses.

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1. Introduction

A trading participant can generate harmful or suspicious orders because of a compromised credential, a software defect, its own order-generation policy, ordinary market conditions, or a combination of these. An operator — the incident-response controller of a broker or trading firm — must often act before the cause is known: throttle, cancel orders, revoke access, or isolate a process. Each action has a cost when applied to a case that did not need it, and each carries risk when withheld from a case that did. The results of this paper concern retrospective diagnosis from complete episodes, with a hypothetical decision cost attached; Section 3.5 states the gap between this and a real-time decision.

Two questions precede any policy question. First: **can the causes be told apart** from the evidence an operator is ordinarily permitted to observe (order acknowledgments, market data, session and process telemetry)? Second, if not: **how costly is that ambiguity**, given that operators must still act? A threshold-based kill switch answers the second question without the first: it trips on an order rate, a cancel ratio or a loss limit and acts, with no attempt to establish cause.

This paper answers the first question with a formal identifiability result (Sections 3–4), scoped to a linear-Gaussian order-flow model, as is the closest structural analog, the literature distinguishing cyberattacks from physical faults in cyber-physical systems. Section 4.1 shows that this confounding is fragile within the linear model itself once the incident propagates through the dynamics, through an exact connection to classical observability. Sections 5–6 check the theory in a linear pilot and define the cost of ambiguity. Section 7 studies, in the ABIDES agent-based market simulator, what matching two incident mechanisms on chosen summary statistics does to classification and decision cost. It is not a test of Theorem 1: the constructions were never shown to satisfy its premise, equality of the full observation-trajectory law.

Contributions.

1. In the linear-Gaussian order-flow model, a compromise and a fault are observationally indistinguishable iff their projected injections coincide

(Theorem 1); fault-mimicking needs the projected compromise direction to lie in the span of the projected fault direction, and rank-two separation gives distinguishability, not high accuracy (Corollaries 1–2, with a Gaussian discriminability benchmark and its unequal-prior ambiguity cost, Proposition 3). Under confounding the cost of ambiguity is bounded below, with a total-variation extension to approximate confounding (Propositions 1–2). A linear pilot is consistent with these predictions (Sections 5–6).

2. Confounding that survives one-shot observation does not survive state feedback unless it lies in the Kalman unobservable subspace; generically it is fragile (Theorem 2, its finite-horizon version Theorem 3, and Corollary 3). In a simulation, an exactly confounded pair becomes classifiable under generic coupled dynamics while it stays at chance under output-level injection.
3. An ABIDES study of observation matching (Section 7). In the four tested injection designs (one simulator, one shared infrastructure), calibration by search did not bring the summary-matched regime to chance, and in some designs the summary-matched regime was more separable than the contrast regime. On the paired Era-3 basis under the current loss convention, the classifier-based policy has a lower point-estimate cost than always-contain in every cell and is never worse than the best constant policy with a supported interval; the cells without a supported improvement over the best constant lie at the low end of the prior grid or in the Study 5 profile, and in one Study 5 cell the best constant is always-contain. Under the rebaselined fault-cost convention the always-contain comparison does not hold in every cell (Section 7.3).
4. A question left to the companion paper (Paper 2) about the cells where the fitted classifier adds no supported value over the best constant policy (Section 7.5).

2. Related work

Decision-theoretic cyber response. Belief-state (POMDP) formulations of cyber defense track an attacker’s progression through a network and choose actions under noisy alerts [1]; related work formulates response as a Stackelberg game over attack-response trees [2], or balances service continuity against intrusion risk under partial observability [3], with learning-based extensions against dynamic attackers and under model misspecification [4, 5],

and model-based decision support [6]. None of this literature prices market-impact costs of the kind Section 6 constructs; our own cost structure is a loss table with a market-impact proxy, not a model of endogenous liquidity replacement.

Cyber-physical fault-versus-attack differentiation. The closest structural analog to our question is the literature distinguishing physical faults from cyberattacks in power and other cyber-physical systems. Pasqualetti, Dörfler and Bullo [7] characterize, with geometric control theory, which attacks are undetectable or unidentifiable by an output-based monitor (those that excite only the zero dynamics); Theorem 2 uses the attack-free special case of this object, the Kalman unobservable subspace. Roy and Dey [8] estimate an anomalous input to a linear system with a sliding-mode unknown-input observer and derive a closed-form distinguishability criterion between fault and attack, including a fault-mimicking condition that is the direct analog of our Corollary 1. Observer-based schemes that detect and isolate faults and deception attacks simultaneously [9], and detectors designed to keep faults and attacks distinguishable in interconnected systems [10], address the design side of the same question; neither prices the response or treats trading. On the data-driven side, Anwar, Mahmood and Shah [11] separate simulated faults from attacks on an IEEE bus test system with a PCA projection and a support-vector classifier, and later systems fuse network and process data for attack-versus-failure attribution [12]. Closest to our decision-theoretic framing, a triage framework for energy cyber-physical systems attributes anomalies to physical faults or to measurement-integrity loss, prices misattribution with response-cost matrices against blanket responses, and observes that exactly matched fault/attack pairs are discriminated at chance [13]. That work is empirical and set in process and wind-farm control; it derives no necessary-and-sufficient indistinguishability condition, no cost floor under confounding and no state-feedback fragility, and does not address trading. Its matched-pair observation is the empirical counterpart of our Theorem 1 and Proposition 1, whereas Section 7 concerns pairs matched only on summary statistics.

Roy and Dey validate their criterion in simulations of the same linear model it is proved for, as does our pilot (Section 5). Section 7 attempts a nonlinear construction in a multi-agent market and does not claim to test their criterion or ours. Stealth that holds in a linear model need not hold in a nonlinear one: stealthy injections must be redesigned for nonlinear differential-algebraic grid models [14]. Alisic and Amin [15] extend sensor-

attack analysis to dynamics on Lie groups and show that an attack keeps its impact and stealth across operating conditions only if it commutes with the flow. No analogous condition has been derived for the order-flow model of Section 3, so we claim neither that our market satisfies it nor that it fails it.

Market manipulation detection and trading kill switches. A large literature detects manipulative patterns (spoofing, wash trading) from trade and order-book data, for example with supervised and unsupervised deep-learning detectors [16]; it classifies trading patterns as manipulative or not and does not ask whether a compromised gateway and a faulty one can be told apart, nor price the response. Manipulative order generation can be automated through compromised brokerage accounts [17], which supports modelling a compromise as a structured perturbation of order flow. A threshold trigger on an order-flow statistic is an observation-dependent policy; the constant policies of this paper (always contain, always leave) are decision-theoretic reference points, the best achievable without any observation, not models of kill switches.

Costly information acquisition. Models of costly information acquisition in asset markets [18] and optimal-stopping formulations of costly sequential acquisition [19] concern traders’ incentives to learn about asset value, not an operator’s decision to verify an incident’s cause; the question matters to the companion paper rather than to this one.

Scope of the contribution. The cyber-physical distinguishability literature [8, 7, 11], cost-aware triage [13] and the Lie-group transfer analysis [15] share substantial machinery with Sections 3–4. What we add is (1) an exact identifiability criterion and cost floor for the compromise-versus-fault question at a trading gateway, with a market-impact loss table rather than a detection-accuracy metric; (2) a comparison, through the unobservable subspace, of output-level and state-level injection of the same incident pair, showing that exact output-level confounding is generically not preserved once the incident propagates (Theorem 2); and (3) an empirical study of whether summary-matching produces confounding in a nonlinear multi-agent simulator. Items (1) and (2) are careful applications of established machinery rather than new theory; (3) is the main empirical contribution. The prior-shift threshold correction of Section 6 is standard [20, 21].

3. Model

Throughout, K (“compromise”) and F (“fault”) name fully specified perturbations of the order-flow process — a directional bias, an elevated order rate or an altered cancellation pattern, with a chosen severity and onset — not claims about real attackers or real defects. The results concern *behavioural-mechanism identification*, whether two specified generative processes can be told apart from gateway-observable order flow, not inference about intent or authorization, which would need other evidence (credential logs, code-change records). The four ABIDES designs of Section 7 are parameterizations of one shared simulation infrastructure, not independent implementations.

3.1. Order-flow dynamics

Let $x_t \in \mathbb{R}^n$ be a latent representation of normal order-flow state (for example, components loading onto price pressure, volume intensity and cancellation propensity), evolving under known linear-Gaussian dynamics with transition matrix A and process-noise covariance Q . The complete model is Definition D1 below.

3.2. Incident injection

At an onset time τ , possibly unknown to the operator, an incident of mechanism $M \in \{K, F\}$ injects an additive perturbation into the pre-observation order-flow signal, with direction b_M and severity θ_M .

Definition D3 (Mechanism instance). A mechanism instance is a triple (M, θ_M, b_M) with cause $M \in \{K, F\}$ (K : compromise, F : fault), a deterministic severity $\theta_M \in \mathbb{R}$ and a deterministic injection direction $b_M \in \mathbb{R}^n$. Statements compare one fixed instance of K with one fixed instance of F (pairwise-instance reading); no distribution over severities, directions or onsets within a cause is modelled.

3.3. Gateway-observable output

The operator observes a lower-dimensional ($m < n$), noisy projection of order flow through a fixed, known matrix $C \in \mathbb{R}^{m \times n}$, representing what a gateway or market-data feed surfaces (spread, depth, cancellation rate). In this model the incident shifts the signal at the point of observation only; Section 4.1 lets it propagate through the dynamics.

Definition D1 (Output-level injection model). Fix n, m , a matrix $A \in \mathbb{R}^{n \times n}$, an observation matrix $C \in \mathbb{R}^{m \times n}$, covariance matrices $Q \in \mathbb{R}^{n \times n}$ and $\Sigma \in \mathbb{R}^{m \times m}$, and an onset time τ . Under a mechanism instance (M, θ_M, b_M) (Definition D3), the latent state, the perturbed state and the observation evolve for $t \geq 0$ as

$$\begin{aligned} x_{t+1} &= Ax_t + w_t, & \tilde{x}_t &= x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, \\ y_t &= C\tilde{x}_t + \varepsilon_t, \end{aligned}$$

with $w_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$. The injection enters the observation only; the state recursion does not see it. (Section 3 additionally takes $m < n$; no statement below uses that restriction.)

Everything except the injection term is common to the two causes. Section 4 rests on the following standing assumptions, which the implementation satisfies; the finite-horizon experiments of Sections 5–6 always include time steps after the onset, so (A5) holds.

Standing assumptions.

(A1) Gaussian, independent noise. $w_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, with Q and Σ symmetric positive semidefinite; the family $\{w_t\}_{t \geq 0} \cup \{\varepsilon_t\}_{t \geq 0}$ is mutually independent (across time and across the two sequences), and its law does not depend on the instance.

(A2) Deterministic initial state. $x_0 = 0$. (The proofs use only that x_0 is Gaussian, possibly degenerate, independent of the noise, with a law that does not depend on the instance; $x_0 = 0$ is the implemented special case.)

(A3) Known, common model. A, Q, C, Σ are known and identical under both causes; only the injection term $\theta_M b_M$ depends on the instance.

(A4) Common onset. The onset τ is deterministic, common to both causes, not necessarily known to the operator. (The proofs use only that τ is deterministic and common; no statement requires a policy to know τ .)

(A5) Window reaches the onset. The observation window W contains at least one $t \geq \tau$. Theorem 2 uses the stronger form $W \supseteq \{t : t \geq \tau\}$ and Theorem 3 the stronger form $W \supseteq \{\tau, \dots, \tau + H - 1\}$.

(A6) Positive-definite observation noise. $\Sigma \succ 0$. Used only to apply Proposition 3 to the model on a finite window: it makes the

covariance of the stacked observations positive definite ($V \succ 0$). No other statement uses a positivity condition on Σ or Q ; Q is only required to be positive semidefinite.

Definition D4 (Observation window; observational indistinguishability). An observation window is a set $W \subseteq \{0, 1, 2, \dots\}$ of time indices, finite or not; the observation trajectory on W is $Y_W = (y_t)_{t \in W}$. Write P_M^W for the law of Y_W under instance (M, θ_M, b_M) (for infinite W , the law on the product σ -algebra, determined by its finite-dimensional marginals). Instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable on W if $P_K^W = P_F^W$. Theorem 1 uses $W = \{t : t \geq 0\}$; “indistinguishable at every $t \geq \tau$ ” (Theorem 2) means $P_K^W = P_F^W$ for $W = \{t : t \geq \tau\}$; Theorem 3 (the finite-horizon version of Theorem 2) uses $W = \{\tau, \dots, \tau + H - 1\}$.

Definition D5 (Distinguishable in the population limit). Two instances are distinguishable in the population limit on W if they are not observationally indistinguishable on W , i.e. $P_K^W \neq P_F^W$ (equivalently $\text{TV}(P_K^W, P_F^W) > 0$, Definition D10). This is a statement about the two laws only; it asserts nothing about the accuracy attainable from finitely many observations.

Definition D11 (Gaussian discriminability). If $Y \mid M \sim \mathcal{N}(\mu_M, V)$, $M \in \{K, F\}$, with a common covariance $V \succ 0$, let $\delta = \mu_K - \mu_F$ and $d = \sqrt{\delta^\top V^{-1} \delta} \geq 0$ (the whitened mean separation).

3.4. The loss table, and what it assumes

The cost analysis uses the two-action, two-cause loss table

Action	Compromise K	Fault F
Contain	C_{OP}	DISRUPT
Leave alone	HARM	0

a special case of the general table below; Proposition 1 keeps these symbols.

Definition D8 (Loss table). The general two-action, two-cause loss table $L(a, M)$, $a \in \{\text{contain, leave alone}\}$, $M \in \{K, F\}$, is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave alone}, K) &= H_K, & L(\text{leave alone}, F) &= H_F, \end{aligned}$$

with $C_K, C_F, H_K, H_F \in \mathbb{R}$. This paper’s table (Section 3) is the special case

$$C_{OP} = C_K, \quad \text{DISRUPT} = C_F, \quad \text{HARM} = H_K, \quad H_F = 0.$$

Propositions 1 and 2 are stated in this paper’s symbols C_{OP} , DISRUPT, HARM.

Definition D9 (Prior, policies, Bayes risk, oracle risk, cost of ambiguity). The cause is drawn as $M = K$ with prior probability $p = P(K)$ and $M = F$ with probability $1 - p$; given M , the observation trajectory Y_W has law P_M^W . A policy is a measurable map π from observation trajectories on W to $[0, 1]$, read as the probability of the action *contain* (randomized policies allowed; a deterministic policy takes values in $\{0, 1\}$). Its expected cost is

$$R(\pi) = p \mathbb{E}_K [\pi(Y_W) C_K + (1 - \pi(Y_W)) H_K] \\ + (1 - p) \mathbb{E}_F [\pi(Y_W) C_F + (1 - \pi(Y_W)) H_F].$$

The best achievable expected cost is $R^* = \inf_{\pi} R(\pi)$; the oracle, which observes M , has expected cost $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$; the cost of ambiguity is $\text{CoA}^* = R^* - R_{\text{oracle}}$. Where the observation is the raw trajectory Y the subscript Y is written (R_Y^* , CoA_Y^*).

Definition D10 (Observation densities; total variation). Let μ be a σ -finite measure dominating P_K and P_F (e.g. the finite measure $P_K + P_F$), and f_K, f_F their densities with respect to μ ; $\int \cdot dy$ denotes integration against μ . The total variation distance is

$$\text{TV}(P, P') = \sup_B |P(B) - P'(B)| = \frac{1}{2} \int |f - f'| dy,$$

so that $1 - \text{TV}(P_K, P_F) = \int \min\{f_K, f_F\} dy$.

The table encodes a scenario in which leaving a fault alone is free, for example a logging error that inflates apparent order counts. The strict value of distinguishing the two causes is zero whenever one action is optimal for both; Proposition 1’s cost floor is a property of loss structures in which the causes call for different actions. The simulated fault mechanisms of Section 7 submit real orders, so Section 6.2 prices a positive fault loss H_F measured on the simulated episodes; with it, every cell has a lower point-estimate cost than always-contain, and the cells without a supported improvement over the best constant policy are listed in Section 7.3. Those cells lie at the low end of the prior grid or in the Study 5 profile. The per-design audit of the fault mechanisms is in Supplement S4.

3.5. Observation timing and scope

Section 1 motivates acting before a diagnosis is possible; the analysis does not match that framing. Theorem 1 and Proposition 1 concern the whole

observation trajectory, and both the linear pilot and the ABIDES study use whole-episode summary features (mean, standard deviation and net drift of each observed channel), not a prefix available at a pre-committed decision deadline. The study is therefore a retrospective diagnosis study: nothing enforces that features use only information available when an operator would have to act. A real-time claim would need an explicit observation cutoff and prefix features; we do not make one. This does not affect the identifiability results, which concern the laws of whole trajectories, but the costs of Sections 6–7 do not measure the value of information at an operational decision point.

4. Theoretical results

All proofs are in Supplement S1, one per statement, each labelled as verified independently or checked by the author.

Theorem 1 (Indistinguishability criterion). Under the model of Section 3, with common onset time τ , mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce identical distributions over the observation trajectory $\{y_t\}_{t \geq 0}$ if and only if

$$\theta_K Cb_K = \theta_F Cb_F. \quad (*)$$

Theorem 1 uses Definition D1, Definition D3, Definition D4 (window $W = \{t : t \geq 0\}$) and assumptions (A1)–(A5).

Corollary 1 (Fault-mimicking compromise — corrected for degenerate cases). Fix $u = Cb_K$, $v = Cb_F$, and a nonzero compromise severity $\theta_K \neq 0$. If the fault severity θ_F may be any real number, then

$$\exists \theta_F \in \mathbb{R} : \theta_K u = \theta_F v \iff u \in \text{span}(v).$$

Corollary 1 uses Definition D3 and condition (*) of Theorem 1; the equivalence itself is linear algebra and uses no probabilistic assumption.

Two precisions. Linear dependence of u and v means rank at most one, and its subcases differ: if $v = 0$ but $u \neq 0$ no θ_F solves the equation, and if both vanish every θ_F does; with positive severities, nonzero u, v must lie on the same ray. For a family of fault directions, choosing one direction gives fault-mimicking when u lies on one of the lines $Cb_F^{(i)}$, while superposing faults gives the weaker condition $u \in \text{span}\{Cb_F^{(i)}\}$; the span form assumes fault superposition is modelled. All results are pairwise-instance statements:

matching one instance of each mechanism says nothing about the laws obtained by averaging over severities or onsets, which would need a mixture model this paper does not specify.

Corollary 2 (Separation implies distinguishability, not high accuracy). If $\text{rank}([Cb_K \ Cb_F]) = 2$, no pair (θ_K, θ_F) other than $(0, 0)$ satisfies (*); for any instance pair with $(\theta_K, \theta_F) \neq (0, 0)$, K and F are then distinguishable in the population limit.

Corollary 2 uses Definition D3, Definition D4, Definition D5 and, through Theorem 1, Definition D1 and assumptions (A1)–(A5).

Separation does not imply good finite-sample classification: the projected directions can be small relative to the noise. The benchmark is the whitened separation d of Definition D11; under equal priors the Bayes-optimal accuracy and AUC are the standard identities [22, 23]

$$\text{Acc}^* = \Phi(d/2), \quad \text{AUC}^* = \Phi(d/\sqrt{2}).$$

A warning on notation. Writing $Z = \phi(Y)$ for the engineered features supplied to a classifier, $R_{\text{oracle}} \leq R_Y^* \leq R_Z^* \leq R_{\text{const}}$, and a feature ablation moves R_Z^* , not R_Y^* . The empirical sections concern R_Z^* ; the two coincide only when ϕ is sufficient, which is not established for any feature set used here.

Proposition 1 (Cost of ambiguity is bounded below under confounding — corrected). The hypotheses $\text{DISRUPT} \neq 0$ and $\text{HARM} \neq C_{OP}$ alone do not give a cost floor: when $\text{HARM} < C_{OP}$, leaving alone is optimal for both causes and the ambiguity cost is zero, so the hypothesis $\text{HARM} > C_{OP}$ is essential for the strict inequality.

Suppose (*) holds (the confounded case), $0 < p < 1$, $\text{DISRUPT} > 0$, and $\text{HARM} > C_{OP}$ (harm strictly worse than routine containment — the assumption the earlier version omitted). Then the best achievable expected cost among policies mapping $\{y_t\}$ to $\{\text{contain, leave alone}\}$ is

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, \ p \text{HARM}\},$$

the oracle’s expected cost is $R_{\text{oracle}} = p C_{OP}$, and

$$\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1 - p) \text{DISRUPT}, \ p(\text{HARM} - C_{OP})\} > 0.$$

Proposition 1 uses Definition D8 (this paper’s symbols), Definition D9 and, through Theorem 1 (to pass from (*) to $P_K^W = P_F^W$), Definition D1, Definition D3, Definition D4 and assumptions (A1)–(A5).

For a general finite loss table $L(a, M)$ the cost of ambiguity is

$$\min_a \{p L(a, K) + (1 - p)L(a, F)\} - p \min_a L(a, K) - (1 - p) \min_a L(a, F),$$

strictly positive exactly when no single action is optimal for both causes at positive probability. Exact equality is a brittle operational target, so we extend the bound to approximate confounding.

Proposition 2 (Total-variation extension of Proposition 1). Let the loss table and prior be as in Proposition 1, with $0 < p < 1$, DISRUPT > 0 and HARM $> C_{OP}$, but do not assume (*). Let f_K, f_F be the observation densities (Definition D10) and set $a = p(\text{HARM} - C_{OP})$, $b = (1 - p)\text{DISRUPT}$. Then

$$\text{CoA}^* = \int \min\{af_K(y), bf_F(y)\} dy = \frac{a + b - \int |af_K - bf_F| dy}{2},$$

$$\min(a, b) (1 - \text{TV}(P_K, P_F)) \leq \text{CoA}^* \leq \min(a, b),$$

and, if a common surrogate law $\bar{P}$ satisfies $\text{TV}(P_K, \bar{P}) \leq \epsilon_K$ and $\text{TV}(P_F, \bar{P}) \leq \epsilon_F$, then

$$\text{CoA}^* \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}.$$

Proposition 2 uses Definition D8 (this paper’s symbols), Definition D9 and Definition D10; it uses none of (A1)–(A6): it holds for any two observation laws.

The first identity is the classical relation between Bayes risk and total variation [24]; the lower bound recovers Proposition 1 when $\text{TV} = 0$. The surrogate-law bound states the premise any “approximate confounding transfers approximately” argument needs: a bound on distributional approximation error, not a match on one statistic. A classifier performing near chance cannot certify a small total variation distance.

For the equal-covariance Gaussian setting, likelihood-ratio integration gives the population-optimal ambiguity cost as a function of d , the unequal-prior cost analog of Corollary 2; it is a standard consequence of Gaussian testing, and a summary-based classifier can only do worse.

Proposition 3 (Gaussian unequal-prior ambiguity cost). Let $Y | M \sim \mathcal{N}(\mu_M, V)$, $M \in \{K, F\}$, with common covariance $V \succ 0$ and discriminability $d > 0$ (Definition D11), prior $p = P(K)$, and the general loss

table with $a = p(H_K - C_K)$, $b = (1 - p)(C_F - H_F)$, $a, b > 0$. Then the population-optimal ambiguity cost is

$$\text{CoA}_Y^*(d) = a \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right) + b \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

which tends to $\min(a, b)$ as $d \rightarrow 0^+$ (the value at $d = 0$, exact confounding, recovering Proposition 1, is obtained as this limit) and decreases toward 0 as $d \rightarrow \infty$.

Proposition 3 uses Definition D8 (general symbols), Definition D9, Definition D10 and Definition D11; applied to the model on a finite window it needs (A1)–(A5) (common covariance) and (A6) ($V \succ 0$).

4.1. Confounding is fragile within the linear model: state feedback

Theorem 1 models the incident as an output-level disturbance: it shifts x_t only at the point of observation, while the recursion $x_{t+1} = Ax_t + w_t$ evolves as if nothing happened. A compromise or a persistent fault behaves more like a state-level disturbance: behaviour at t feeds into the state at $t + 1$, as an incident agent's orders feed into the book that other agents react to. This changes the confounding condition.

Definition D2 (State-level injection model). As Definition D1, except that the perturbed state drives the recursion from the onset on:

$$\begin{aligned} x_{t+1} &= Ax_t + w_t \quad (t < \tau), & x_{t+1} &= A\tilde{x}_t + w_t \quad (t \geq \tau), \\ \tilde{x}_t &= x_t + \mathbb{1}[t \geq \tau] \theta_M b_M, & y_t &= C\tilde{x}_t + \varepsilon_t. \end{aligned}$$

The injection is persistent (added at every $t \geq \tau$) and propagates through A .

Definition D6 (Unobservable subspace; observability). The (Kalman) unobservable subspace of (A, C) is $\mathcal{N}(A, C) = \bigcap_{k=0}^{n-1} \ker(CA^k)$. The pair (A, C) is observable if $\mathcal{N}(A, C) = \{0\}$, equivalently if the stacked matrix

$$\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has rank n .

Theorem 2 (State-feedback confounding requires unobservability). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$. Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable at every $t \geq \tau$ if and only if Δ_0 lies in the unobservable subspace of the pair (A, C) , i.e.

$$\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k),$$

the classical (Kalman) unobservable subspace of (A, C) , where n is the state dimension.

Theorem 2 uses Definition D2, Definition D3, Definition D4 (window $W = \{t : t \geq \tau\}$), Definition D6 and assumptions (A1)–(A4) with (A5) in its all-time form.

Theorem 3 (Finite-horizon version of Theorem 2). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$, let $H \geq 1$ be the number of post-onset observations, and define

$$G_H = \begin{bmatrix} C \\ C(I + A) \\ \vdots \\ C \sum_{j=0}^{H-1} A^j \end{bmatrix}.$$

Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce the same law of the observations over the window of H post-onset observations if and only if $G_H \Delta_0 = 0$, which holds if and only if $CA^j \Delta_0 = 0$ for $j = 0, \dots, H-1$. If $H \geq n$, this is equivalent to the all-time condition of Theorem 2, $\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k)$.

Theorem 3 uses Definition D2, Definition D3, Definition D4 (window $W = \{\tau, \dots, \tau + H - 1\}$), Definition D6 and assumptions (A1)–(A4) with (A5) in its finite-horizon form.

The result is not inherently asymptotic: for $H \geq n$, exact equality over H post-onset observations already forces the all-time Kalman condition. What depends on horizon and noise is finite-sample discriminability when the condition fails, which Theorem 2 does not bound.

Corollary 3 (State-feedback confounding under observability). Under the hypotheses of Theorem 2, if (A, C) is observable, then mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable at every $t \geq \tau$ if and only if $\theta_K b_K = \theta_F b_F$.

Corollary 3 uses everything Theorem 2 uses, and Definition D6; its accompanying remark also uses Definition D7.

Definition D7 (Generic). For fixed C , a property of A holds generically if it holds for Lebesgue-almost every $A \in \mathbb{R}^{n \times n}$, i.e. it fails only on a Lebesgue-null subset of $\mathbb{R}^{n \times n}$.

Remark (Theorem 1’s confounded pairs are generically state-feedback-fragile). Theorem 1 only requires $\Delta_0 \in \ker(C)$ — a single condition. Theorem 2 requires Δ_0 to survive n such conditions simultaneously ($\ker(C), \ker(CA), \dots, \ker(CA^{n-1})$). Whenever (A, C) is observable (for fixed $C \neq 0$, the generic case for a randomly chosen A), the unobservable subspace is $\{0\}$, so **the only state-feedback-robust confounding is the trivial one where $\theta_K b_K = \theta_F b_F$ exactly as vectors** — i.e., the two mechanisms must push the identical linear combination of latent state variables, not merely produce the identical *projection* under C .

The pilot’s construction (Section 5), which uses the null space of C to make $b_K \neq b_F$ with $Cb_K = Cb_F$, is exactly such a pair: $\Delta_0 \in \ker(C)$ but $\theta_K b_K \neq \theta_F b_F$. By Corollary 3 it cannot remain confounded under state feedback whenever (A, C) is observable, which for fixed $C \neq 0$ holds for Lebesgue-almost every A . With the pilot’s A , a scalar multiple of the identity, $\ker(C)$ is trivially A -invariant, a measure-zero case in which the pair stays confounded. We re-ran the pilot with a stable, coupled and observable A . Under state-level injection with generic coupled dynamics, classification accuracy rises with injection magnitude, while under output-level injection it stays at chance throughout; with scalar (uncoupled) dynamics the state-level case also stays at chance. These classifiers use whole-episode features; how quickly the difference becomes detectable after onset is not measured here (Supplement S2).

Two qualifications. “Generic” concerns a full-measure set of matrices; real, structured market dynamics can have unobservable components by construction, and we have not established that they behave like a generic matrix. A one-time initial-state perturbation and persistent forcing produce different mean-difference sequences but share the same unobservability condition, so persistence is not what makes confounding fragile; state-level injection is. Supplement S2 gives two further qualifications (finite-horizon discriminability; identical state effects with different meanings).

For Section 7 this gives a purely linear reason why constructed confounding may not survive in a market, where an incident’s orders propagate through the book. It is a mechanism proved within the linear model, not one

isolated in the nonlinear simulator (Section 7.4).

5. Pilot-scale validation in the linear model

We instantiate the model at small scale (a few latent dimensions, fewer observed channels, AR(1) dynamics; parameters in Supplement S2) as a check that the theory is internally consistent and detectable in simulation, not as a market-realistic study. Two regimes are built by the choice of injection directions: *confounded*, with b_K, b_F chosen through the null space of C so that $Cb_K = Cb_F$ exactly, satisfying Theorem 1’s condition; and *separated*, with Cb_K, Cb_F linearly independent. A logistic-regression classifier is trained on whole-episode summary statistics of each observed channel (mean, standard deviation, net drift), in both regimes and at two injection magnitudes.

Table 1: Linear pilot: held-out accuracy and AUC by regime and injection magnitude (480 test episodes per cell).

Regime (Theorem 1’s prediction)	Magnitude	Accuracy	AUC
Confounded (indistinguishable)	0.6	0.529	0.519
Separated (distinguishable)	0.6	0.938	0.983
Confounded (indistinguishable)	1.2	0.487	0.485
Separated (distinguishable)	1.2	1.000	1.000

As Theorem 1 predicts, accuracy in the confounded regime sits at chance regardless of injection magnitude: a more severe incident is not easier to classify when the two observation laws are identical. In the separated regime accuracy is high and improves with magnitude, as Corollary 2 predicts through the whitened separation. Three further classifier families (a random forest, Gaussian naive Bayes and a single-feature threshold rule) give the same pattern (Supplement S2), as Theorem 1 implies for any passive classifier.

6. Cost of ambiguity: definitions, estimator and the linear pilot

Terminology. Three risks are kept separate, in the order $R_{\text{oracle}} \leq R_Y^* \leq R_{\text{const}}$: the true-cause oracle $R_{\text{oracle}} = \mathbb{E}[\min_a L(a, M)]$, the best policy using the observation $R_Y^* = \inf_{\pi} \mathbb{E}[L(\pi(Y), M)]$, and the best constant action $R_{\text{const}} = \min_a \mathbb{E}[L(a, M)]$. Exact confounding is sufficient, not necessary, for $R_Y^* = R_{\text{const}}$: an informative Y can still leave the optimal action unchanged,

for example at a low prior where every posterior stays below the containment threshold. An absence of measured cost benefit is therefore not evidence that two observation laws are equal.

Prior-shift correction of the threshold. The classifier is trained on balanced data, so its score $s(y)$ estimates $P(K | Y = y)$ under the training prior $p_{\text{train}} = \frac{1}{2}$, not the deployment prior p . Applying the deployment threshold $q^* = \text{DISRUPT}/(\text{HARM} - C_{OP} + \text{DISRUPT})$ to this raw score is the prior-shift error documented for cost-sensitive learning [20, Sec. 3, Thm. 2]; see also [21]. We apply Bayes’ rule, $q_p(y) = \frac{(p/p_{\text{train}})s(y)}{(p/p_{\text{train}})s(y) + \frac{1-p}{1-p_{\text{train}}}(1-s(y))}$, and threshold q_p at q^* . Without the correction, a classifier under confounding appears worse than ignoring the observation, an artifact of the threshold rather than of confounding. The correction trades false positives for false negatives rather than removing error: at low priors it makes the policy conservative about calling a compromise (Supplement S4). It is more accurate about cost for a given loss table, not uniformly more accurate.

Linear pilot. With an illustrative toy triple $(C_{OP}, \text{DISRUPT}, \text{HARM})$ satisfying Proposition 1’s hypotheses (Supplement S2), we compare four policies across a grid of priors $P(K)$: always contain, always leave, the prior-corrected classifier, and the oracle.

Table 2: Linear pilot, confounded regime: expected cost per episode (toy units). Under exact confounding $R_Y^* = R_{\text{const}}$, so the last column is the classifier’s excess over the best achievable cost: mean over seeds, with the range over seeds in brackets. Per-seed values are test-set estimates and can fall on either side of the floor.

$P(K)$	Contain	Leave	Classifier (sd)	Oracle	R_{const}	Classifier $- R_{\text{const}}$ [min, max]
0.05	5.850	0.500	0.500 (0.000)	0.150	0.500	0.000 [0.000, 0.000]
0.1	5.700	1.000	1.000 (0.000)	0.300	1.000	0.000 [0.000, 0.000]
0.2	5.400	2.000	2.000 (0.000)	0.600	2.000	0.000 [0.000, 0.000]
0.3	5.100	3.000	3.000 (0.000)	0.900	3.000	0.000 [0.000, 0.000]
0.5	4.500	5.000	4.520 (0.085)	1.500	4.500	0.020 [-0.125, 0.196]

The constant and oracle policies match their closed forms $pC_{OP} + (1 - p)\text{DISRUPT}$, $p\text{HARM}$ and pC_{OP} , a direct check of the cost computation. Under confounding no policy of Y has expected cost below $R_Y^* = R_{\text{const}}$, so the fitted classifier can at best reach this floor, and it did so exactly at every prior below $q^* = 0.462$: in all 20 seeds it took the best constant action on every test episode. At the highest prior, the only one above q^* and the closest to it, the fitted score straddles the threshold and the classifier left

some episodes uncontained; its cost then scatters around the floor (mean excess 0.020, seed standard deviation 0.085; 9 of 20 seeds at or below it), and seeds below the floor reflect finite test sets, not a gain. Proposition 1 concerns the Bayes-optimal policy; it implies that a fitted classifier cannot beat the floor, not that it attains it. Under separation the classifier beats the best constant policy (Supplement S2, with a sensitivity analysis over cost triples).

6.1. Where the dollar cost triple comes from

For the ABIDES study we reconstruct a dollar cost triple $(C_K, C_F, H_K) = (C_{OP}, \text{DISRUPT}, \text{HARM})$ from the Almgren–Chriss market-impact model [25], $L(X, T) = \varepsilon X + (\frac{\gamma}{2} + \frac{\eta}{T})X^2$, with the model’s own illustrative parameters applied to three scenarios: a routine containment trade, a benign order forced into an urgent unwind, and an unmitigated adversarial sequence. This gives $(C_K, C_F, H_K) = (\$490,938, \$473,958, \$1,317,500)$, which satisfies $H_K > C_K$. It is an arithmetic illustration, not a calibration to any venue or incident; the costs are not measured economic losses (worked example in Supplement S4).

6.2. A positive fault loss

The simulated fault mechanisms submit real orders, so leaving a fault alone is not free. We replace the zero entry by H_F , the Almgren–Chriss cost of the fault mechanism’s directly measured submitted volume in each episode, averaged over episodes (so the convexity of L in X is included); it is a modelled cost of submitted, not executed, volume, measured on the same episodes as the classification cells (Section 7.4). Per-design values are in Supplement S4; they range from \$850 to \$104,739 per episode. With $H_F > 0$,

$$\begin{aligned} R_{\text{oracle}} &= p C_K + (1 - p) H_F, \\ R(\pi) - R_{\text{oracle}} &= p (H_K - C_K) \text{FNR} + (1 - p) (C_F - H_F) \text{FPR}, \end{aligned}$$

and the cost-optimal threshold on the prior-corrected score is $q^* = (C_F - H_F) / [(H_K - C_K) + (C_F - H_F)]$, which depends on the loss table only (between 0.309 and 0.364 across designs and regimes). The zero- H_F formulas above hold for the toy scenario only. Each cell’s cost is estimated by direct class-conditional weighting, $\hat{R}_p = p\{C_K + (H_K - C_K)\text{FNR}\} + (1 - p)\{H_F + (C_F - H_F)\text{FPR}\}$, from the held-out error rates at q^* , with no resampling step. A

sensitivity arm uses an alternative convention in which both fault entries are absolute.

Consequence for the policy comparison: per the full 50-cell audit (Section 7.3, using the direct class-conditional weighting method), the classifier has a lower point-estimate cost than always-contain in all 50 cells (a supported improvement in 49) and is never worse than the best constant policy with a supported interval; its advantage over the best constant policy is a supported improvement in 40 cells, and the cells without one lie at the low end of the prior grid or in the Study 5 profile (Section 7.3).

7. Observation matching in an agent-based market

7.1. Scope and constructions

What this section does and does not test. Theorem 1 concerns equality of the whole observation-trajectory law, $P_K(Y_{0:T}) = P_F(Y_{0:T})$. The constructions below match the two mechanisms on one or a few chosen summary statistics while other observable channels may still differ. Matching one moment is not equality of the joint law, and a classifier exploiting an unmatched channel does not contradict Theorem 1, even in a linear system: $\mathcal{N}((0, 2), I)$ and $\mathcal{N}((0, -2), I)$ share their first-coordinate mean. No linearization bridge (operating point, Jacobian, approximation-error bound) connects the linear model to the simulator. This section is therefore not a test of Theorem 1; it asks which observable channels leak information under which summary-matching choices, and what that does to decision cost.

We inject compromise and fault mechanisms as order attributes (submission rate, order size, price aggressiveness, directional bias, cancel/replace churn) into ABIDES [26], a discrete-event multi-agent market simulator with a background population of noise, value and market-maker agents. Each design has two regimes. In the *summary-matched* regime a calibration search sets the fault’s attributes so that it matches the compromise on a chosen target statistic; in the *contrast* regime the two mechanisms are deliberately given a large difference in a chosen attribute. The classifier (logistic regression) uses whole-episode summaries of the observed channels (Section 3.5).

Five profiles. The *original design* (V1) varies all five attributes and calibrates on mean price drift, with four observed channels (spread, depth, cancellation rate, mid-price). *Study 4* reuses the original design’s episodes with three added channels: order-flow imbalance [27], signed executed volume and queue concentration. *Study 5* (V2) varies a different attribute subset

with a deterministic wake-up clock and a different calibration target. The *third design* (V3) calibrates on total order count and leaves the cancellation rate free. The *fourth design* (V4) varies directional bias, size and aggressiveness and calibrates on mean spread. The designs differ in which attributes are varied and which statistic is matched, the choices that determine what can leak. We stopped adding designs when the first three had shown three distinct failure modes; a fourth, built to probe that stopping rule, produced a further distinct profile, so the design space is not exhausted (Supplement S3).

7.2. Classification on the paired basis

Table 3: Classification by design and regime. The ten classification cells are computed on paired seeds: each seed yields one compromise and one fault episode, the train/test split is grouped by seed, and the intervals are cluster-bootstrap intervals over seeds, all under the hardened pipeline (Section 7.4). V1: original design; Study4: V1 episodes with the expanded channel set; V2: Study 5; V3: third design; V4: fourth design. [†] Zero test errors: exact Clopper–Pearson interval over the test episodes (the bootstrap interval is degenerate; Supplement S6).

Design	Contrast		Summary-matched	
	Accuracy [95% CI]	AUC [95% CI]	Accuracy [95% CI]	AUC [95% CI]
V1_original	0.685 [0.652, 0.718]	0.756 [0.728, 0.786]	0.857 [0.830, 0.882]	0.934 [0.917, 0.950]
Study4_expanded21ch	0.998 [0.995, 1.000]	1.000 [1.000, 1.000]	0.898 [0.875, 0.920]	0.966 [0.954, 0.977]
V2_study5	0.663 [0.633, 0.692]	0.731 [0.706, 0.757]	0.553 [0.525, 0.580]	0.577 [0.551, 0.604]
V3_third	0.998 [0.995, 1.000]	1.000 [1.000, 1.000]	0.875 [0.850, 0.898]	0.953 [0.940, 0.967]
V4_fourth	0.727 [0.698, 0.757]	0.807 [0.783, 0.832]	1.000 [0.994, 1.000] [†]	1.000 [1.000, 1.000]

Every cell has 1000 episodes per class and 600 held-out test episodes from 300 test seeds. A cell is *at chance* when its accuracy interval contains one half, and *above chance* when the lower end of its accuracy interval is above one half; the interval is the cluster bootstrap over test seeds, or the exact Clopper–Pearson interval for a zero-error cell. An interval containing one half would not establish equal observation laws: a wide interval leaves room for undetected signal.

Table 3 shows no single ranking. In the original design the summary-matched regime is the more classifiable one, with non-overlapping accuracy intervals; with Study 4’s added channels the contrast regime becomes nearly perfectly classifiable, because the contrast mechanisms were always separable on signed executed volume, which the original four channels did not expose (Supplement S3). Study 5 is weakly classifiable in both regimes. In the third design both regimes are strongly classifiable: its mechanisms differ in cancellation rate, one of the observed channels, which matching total order count

left free. The fourth design’s summary-matched cell has 0 test errors (exact 95% Clopper–Pearson lower bound 0.994 over test episodes, 0.988 over test seeds): calibrating on mean spread left a large difference in effective order size, which leaks through depth: a size-based channel, not the cancellation-rate channel of the third design. In each of the four injection designs, calibration by search left the summary-matched regime classifiable above chance on the paired Era-3 basis (Table 3: the lower end of every summary-matched accuracy interval lies above one half, the lowest being Study 5’s, 0.525); for Study 5 the residual signal is weak (0.553 accuracy, interval 0.525–0.580). The four designs are parameterizations of one shared simulation infrastructure (Section 9), so this is a regularity of the constructions tested, not a general law about calibration by search.

The mechanism is structural. In the linear model one vector equality in $\mathbb{R}^m$ removes the difference between mechanisms in every observed coordinate at every time step, because C is fixed and known, and even that equality is fragile under state feedback (Theorem 2). In a multi-agent market there is no fixed linear map to match: a designer matches a finite set of statistics, and whatever is left unmatched can leak. How much leaks depends on the observer’s channels (the rows of C in the linear analog), on the incident’s footprint relative to background noise, and on classifier capacity — all representable, though unexplored, in the linear pilot. What the linear model does not need is calibration by search: its confounded pair has a closed-form construction, whereas the simulator’s map from attributes to statistics has none. None of the summary-matched constructions was matched to a pre-specified tolerance on independent data (Supplement S3), so “summary-matched” means a match attempted, not achieved.

7.3. Cost of ambiguity on the paired basis

The cost table applies the dollar triple of Section 6.1 and the measured H_F of Section 6.2 to five profiles $\times$ two regimes $\times$ five priors (50 cells; full table in Supplement S4). Study 4 shares its incident mechanism with the original design and reuses its H_F . The best constant policy’s excess over the oracle agrees with the identity $B(p) = \min\{p(H_K - C_K), (1 - p)(C_F - H_F)\}$ in every cell. Intervals on differences between policies are cluster-bootstrap intervals over test seeds; the constant policies’ costs are not resampled.

At point estimate (strict inequality), the classifier beats always-contain in 50 of 50 cells and the best constant policy in 44 of 50. With cluster-bootstrap

intervals over seeds, its advantage over always-contain is a supported improvement in 49 cells and inconclusive at this power in 1; against the best constant it is a supported improvement in 40 cells and inconclusive in 10. No cell shows a supported deterioration against either comparator. The cells without a supported improvement are listed in Table 4. These cells lie at the low end of the prior grid or in the Study 5 profile; in all of them the best constant policy is always-leave, except the Study 5 summary-matched cell at the highest prior, where it is always-contain. Point-estimate counts use strict inequality; Table 4 gives the interval verdicts.

Table 4: Cells whose advantage over the best constant policy is not a supported improvement (paired basis, current fault-cost convention). Costs in dollars (illustrative reconstruction, Section 6.1); difference = classifier minus best constant, with its cluster-bootstrap 95% interval over seeds.

Design	Regime	Prior	R_{clf}	Best constant	Difference [95% CI]	vs best constant	vs always-contain
V1_original	contrast	0.05	99,906	97,510 (leave)	2,395 [−1,791, 7,564]	inconclusive	improvement
V1_original	contrast	0.10	160,009	161,720 (leave)	−1,711 [−7,827, 5,451]	inconclusive	improvement
V2_study5	contrast	0.05	67,767	66,682 (leave)	1,085 [−827, 4,357]	inconclusive	improvement
V2_study5	contrast	0.10	128,699	132,515 (leave)	−3,816 [−6,894, 401]	inconclusive	improvement
V2_study5	summary-matched	0.05	66,682	66,682 (leave)	0 [0, 0]	inconclusive	improvement
V2_study5	summary-matched	0.10	132,515	132,515 (leave)	0 [0, 0]	inconclusive	improvement
V2_study5	summary-matched	0.20	266,863	264,180 (leave)	2,683 [−551, 7,178]	inconclusive	improvement
V2_study5	summary-matched	0.30	400,582	395,845 (leave)	4,737 [−6,297, 15,234]	inconclusive	improvement
V2_study5	summary-matched	0.50	481,261	482,448 (contain)	−1,187 [−6,308, 4,133]	inconclusive	inconclusive
V4_fourth	contrast	0.05	69,353	71,486 (leave)	−2,133 [−5,648, 2,486]	inconclusive	improvement

The exceptions are concentrated in Study 5 (5 of the 6 point-estimate exceptions); the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 4). At the lowest priors of the Study 5 summary-matched profile the classifier’s cost equals the best constant’s exactly: a tie, counted as an exception under strict inequality. Every Study 4 contrast cell is a supported improvement, whereas the original design’s lowest-prior contrast cells are inconclusive. Under the alternative fault-cost convention (both fault entries absolute), the classifier beats always-contain at point estimate in 39 of 50 cells, and no cell shows a supported deterioration against either comparator; the choice of convention is stated in Section 6.2.

7.4. Checks that bound the claims

Negative control. Two incident variants with identical observable order-generation kernels, differing only in a behaviourally inert label, were run through the full pipeline, with compromise and fault seeds interleaved from one pool and independent replicate datasets rather than resplits of one dataset.

With identical kernels, the tested classifier shows no detectable accuracy signal (between-replicate interval inside the pre-specified equivalence margin); AUC does not meet the margin. The control checks the pipeline and evaluation boundary, not Theorem 1. No formal two-one-sided-tests procedure was run (intervals and margin in Supplement S5).

Provenance of each result. The ten headline classification cells (Table 3) and the cost table built on them were generated with the hardened pipeline (spawn, one task per worker, runtime patch fingerprint passing for every task), on paired seeds. The measured fault-side harm H_F used by the cost table is computed per fault episode from those same episodes, whose trajectories were recovered and checked bit-identical to the stored features, not from an earlier run. Every other ABIDES result carries its generation era (Supplement S6); results from the two earlier eras were produced before the RNG patches or with scripts that would not detect a mid-run patch, are reported as exploratory, and not all of their seed blocks are in the provenance register.

Persistence ablation (exploratory). In one run per condition, made before the hardened pipeline, removing persistence lowered classifiability in both scenarios, by a larger margin in the summary-matched scenario than in the contrast scenario, consistent with the duration and dose confound of that design; we do not attribute the difference to any isolated mechanism, and it is not a test of Theorem 2. Supplement S5 describes it, with the isolation redesign (inconclusive), leave-one-design-out on the paired basis (the separating direction does not transfer across designs; a robustness check, not a claim), the signed-volume convention and the error-rate decomposition; Supplement S3 reports the earlier-era secondary studies (seed and classifier-family robustness, population scale, per-feature isolation, footprint sweep).

7.5. *A question left to Paper 2*

The cells of Section 7.3’s cost table in which the fitted classifier’s advantage over the best constant policy is not a supported improvement lie at the low end of the prior grid or in the Study 5 profile (Table 4): passive observation adds no supported value there. Whether an action on a different information channel can add value in those cells is outside this paper. The companion paper (Paper 2) asks, for the cells of this set in which the classifier’s point estimate does not beat the best constant policy, whether targeted verification does, and over which region of the verification parameters. It examines this as an exploratory test. This paper claims no outcome for

that test, and does not claim that these cells are where verification’s effect is largest.

Two results bound what this paper contributes to that question. Whether K and F are confounded (Theorem 1) is a property of the mechanism instances and the observation law, not of the prior p ; what depends on p is the value of resolving the ambiguity, Proposition 1’s $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$, zero exactly when $p \in \{0, 1\}$. And exact confounding is not generic under state-level injection (Theorem 2), while no ABIDES construction was shown to instantiate Theorem 1’s premise. This paper therefore defines no confounded region for a real incident and evaluates no verification channel.

8. Discussion

The linear identifiability boundary is exact within its scope and remains a legitimate diagnostic wherever that scope holds, plausibly small perturbations around a steady-state book. Its scope is narrower than Theorem 1 alone suggests. Confounding that survives an output-level check need not survive feedback into the dynamics, and the gap is governed by a classical object, the unobservable subspace of (A, C) (Section 4.1). The ABIDES study adds a second, independent caution: engineering a nonlinear analog of exact confounding by matching summary statistics left every summary-matched regime classifiable above chance, through channels that depended on the design. That calibration needed a search is a fact about these two constructions, not a dividing line between linear and nonlinear systems. A third caution is methodological: a decision-cost result is only as reliable as the decision rule behind it, and a threshold set without prior-shift correction can manufacture a cost finding in either direction (Section 6). For practitioners, Supplement S7 turns these observations into a checklist, restricted to what the paired-basis results and the audits support.

9. Limitations

1. **Retrospective, whole-episode diagnosis.** Every classifier uses complete-episode features; no result measures the value of acting at a real decision deadline (Section 3.5).
2. **One simulator, one shared infrastructure.** The four ABIDES designs are parameterizations of one shared code base (configuration

builder, kernel, features, cost pipeline), not independently coded implementations, so a bug common to that infrastructure would not be caught by comparing designs. The five profiles are regularities of the constructions tested; the design space is open-ended.

3. **Illustrative costs.** The dollar triple is an Almgren–Chriss illustration, not a calibration, and H_F prices submitted, not executed, volume with a market-impact model; neither is a measured economic loss.
4. **Reference policies.** The constant policies are decision-theoretic reference points, not models of deployed kill switches; an operational comparison would need a specified threshold detector with comparable feature access and timing.
5. **Power.** The cells listed in Table 4 are inconclusive at the sample size used, not shown to be ties; and the persistence ablation is exploratory, one run per condition before the hardened pipeline (Section 7.4).

10. Conclusion

For a linear-Gaussian model of gateway-observable order flow, a compromise and a fault are observationally indistinguishable if and only if their projected injections coincide, and under that confounding no policy using the observation beats the best constant action; the bound extends to approximate confounding in total variation.

Within the same linear model, this confounding is fragile: once the incident propagates through the dynamics, it survives only when the injection difference lies in the Kalman unobservable subspace, which is trivial for generic dynamics.

In the ABIDES simulator, calibration by search did not bring any summary-matched regime to chance, and which channel leaked depended on the design; the study is about observation matching, not a test of the theorem.

With a measured positive fault loss (Section 6.2): on the paired Era-3 basis under the current loss convention, the classifier-based policy has a lower point-estimate cost than always-contain in every cell and is never worse than the best constant policy with a supported interval; the cells without a supported improvement over the best constant lie at the low end of the prior grid or in the Study 5 profile, and in one Study 5 cell the best constant is always-contain. Under the rebaselined fault-cost convention the always-contain comparison does not hold in every cell. The cells in which the fitted classifier adds no supported value over the best constant policy (Section 7.3)

leave a question to the companion paper (Paper 2), stated in Section 7.5 together with what this paper does not claim about it.

Data and code availability

The simulation code, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). Until then they are available from the corresponding author on reasonable request. The ABIDES market simulator is open-source software; the patches applied to it are part of the deposit.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

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